

Friday, February 20, 2026

Office of the Deputy Secretary and Assistant Secretary for Technology Policy
Office of the National Coordinator for Health Information Technology
U.S. Department of Health and Human Services
200 Independence Avenue, SW
Washington, DC 20201

Re: Request for Information: Accelerating Adoption and Use of AI in Clinical Care

ECRI is submitting the following comments in response to HHS's Request for Information (RFI) on accelerating the adoption and use of artificial intelligence (AI) in clinical care. As an independent nonprofit organization advancing evidence-based practices to improve patient safety, healthcare quality, and cost-effectiveness, ECRI supports efforts to ensure that AI tools used in the clinical environment are safe, effective, transparent, and worthy of public trust.

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The greatest barriers to AI innovation and adoption are not algorithmic limitations, but gaps in real-world readiness, including:

- Procurement processes that do not reward transparency or safety
- Limited independent validation
- Fragmented data systems
- Poor interoperability
- Post-implementation monitoring
- Cybersecurity and reliability vulnerabilities
- Unclear accountability when AI influences care

Highly publicized failures of widely deployed prediction tools demonstrate that AI adoption without evidence can weaken trust and stall progress. One example was seen during [external validation of EPIC's AI-enabled sepsis prediction program](#). America gains the most when innovation is tested, proven, and trusted, not when shortcomings are discovered post-deployment.

A recent [analysis of ways to improve adoption of digital technologies](#), including AI, highlights that creating clear reimbursement pathways and integrating AI tools into clinical practice guidelines are essential adoption enablers.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

HHS can strengthen adoption of clinically valuable AI by prioritizing actions that reward transparency, accountability, and sustained performance. ECRI encourages HHS to:

- Communicate, support and enforce expectations for clinical AI performance and accountability.
- Expand reimbursement pathways for validated clinical AI interventions.
- Promote transparency standards that reward vendors who clearly communicate capabilities, limitations, and performance.
- Clarify expectations for managing AI updates and performance drift over time.

These measures do not restrict innovation—they differentiate serious innovators from opportunistic ones.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Non-medical AI tools—such as general-purpose large language model (LLMs) chatbots, create a distinct governance gap. They can influence clinical decisions and patient behavior, but they often sit outside the accountability, validation, and lifecycle controls that healthcare organizations expect for clinical technology.

Some of the unique legal and implementation issues include:

- **Accountability Gaps**
 - **Diffused liability:** Vendor disclaims clinical use and health system assumes tool is “just informational” but clinicians use it for clinically actionable queries. When harm occurs, responsibility is unclear.
 - **Indemnification mismatches:** Contracts often lack meaningful indemnity for clinical harm, hallucinated content, or latent model defects. When harm occurs, no one is responsible and improvements are not stimulated as a result.
- **Unvalidated “Clinical Advice”**
 - LLM outputs can be very convincing even when inaccurate, which creates automation bias and subtle influence even when clinicians think they are not being influenced.
- **Inappropriate Generalization**
 - Performance varies by context, population, language, and clinical domain. Without transparent limitations, LLMs can amplify disparities or mislead in edge cases.

- **Update Opacity and Performance drift**
 - Unannounced model changes (or retrieval/source changes) can alter behavior without warning, undermining clinical governance, training, and auditability.
 - Lack of stable versioning makes incident investigation difficult: “What would the model have said on the date of the incident?” may be unknowable.
- **Traceability, audit, and record integrity**
 - Chat interactions may not be logged in a clinically meaningful way, or logs are stored outside the organization’s control.
- **Privacy, security, and secondary use**
 - Prompt content may include Protected Health Information (PHI), sometimes inadvertently. Ambiguity around data retention, model training on user data, and potential for data transfer to countries outside the U.S. as part of downstream processing complicates compliance and risk.
 - Tool integrations (browser, plugins, AI agents) expand the attack surface and create new routes for data exfiltration or manipulation potentially in the form of cybersecurity risks like prompt injection and tool hijacking.

What role should HHS play? Here are recommended actions:

Set enforceable expectations for clinical-impact AI, regardless of device status. HHS should clarify that if a system is used in ways that materially influence clinical decisions, it should meet baseline expectations for:

- **Transparency** (intended use, verification/validation test methods, limitations, known failure modes)
- **Traceability** (versioning, audit logs, data provenance)
- **Lifecycle governance** (change management, monitoring, incident reporting)
- **Truthful performance claims** (accurate performance metrics across populations and sub populations)

Use payment and program levers to incentivize governance. Through Centers for Medicare & Medicaid Services (CMS) programs, HHS could condition participation or reimbursement on healthcare organizations demonstrating that they:

- Maintain an AI inventory (including non-device tools such as LLMs).
- Apply risk-tiered evaluation (piloting) prior to use.
- Implement ongoing monitoring and user training.
- Document when and how AI is used in care processes.
- Have an error reporting program in place.

Provide standard procurement and contracting language. Model contract clauses for health systems should cover:

- Minimum logging/audit requirements.
- Versioning and advance notice of model updates.
- Data use and retention limits (including “no training on our data” unless explicit).
- Vendor obligations for safety incidents and vulnerability disclosure.
- Realistic indemnification aligned to risk.

Clarify privacy/security guardrails for LLM workflows. Through the HHS Office for Civil Rights and the Office for the National Coordinator for Health IT (ONC), HHS can issue practical guidance on:

- What constitutes acceptable handling of PHI in prompts.
- HIPAA compliant Business Associate Agreements (BAAs) and subcontractor chains for LLM vendors who could encounter PHI.
- Minimum security controls for AI-enabled programs with the ability to influence patient care, i.e. those connected to electronic health records (EHRs).
- Safe defaults (redaction, local processing options, retention minimization).

Create an enforceable “truth-in-claims” regime for AI with clinical capabilities. HHS can coordinate with the Federal Trade Commission (FTC) and Food and Drug Administration (FDA) so vendors cannot imply clinical-grade abilities without:

- Disclosing evaluation methods.
- Specifying populations/settings tested.
- Accurately reporting performance metrics.
- Reporting/disclosing known limitations and uncertainty.

Non-medical devices that use AI can shape clinical decisions at scale while escaping the normal checks that protect patients. HHS does not need to micromanage innovation, but it should communicate expectations, support adoption with practical guardrails, and enforce accountability so the systems used in care are measurable, monitorable, and honest about what they are not able to do in a way that end users can understand. Strong accountability builds confidence for clinicians, patients, and investors alike.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

ECRI believes that HHS should support a “prove-it-first, monitor-it-always” approach for any AI tool capable of influencing patient care. Before deployment, HHS should require the following information from vendors whose non-medical AI-enabled device has the ability to influence patient care. “Having the ability to” is defined as capable of performing in that area without sufficient design controls to limit risk to use.

Pre-Deployment Information:

1. Intended-Use, task-based validation (rather than general model quality): Evaluate the system on specific clinical tasks it will support and include realistic environment features such as representative users, settings, and EHR-integrated contexts, as appropriate.
2. Reference-grounded testing: Explicitly test convincing but wrong inputs and report whether the model agrees or corrects them in the output. If the device employs applicable features such as LLMs to summarize guidelines or literature, require source traceability and test citation fidelity as part of the verification/validation.
3. Human factors and workflow simulation: Conduct usability tests in simulated clinical scenarios (time pressure, interruptions).
4. Safety-focused red teaming: Structured red-team suites for hallucinations (fabricated facts, contraindications), unsafe omissions (missed warnings, missing follow-up), over-confident tone, prompt injection/tool hijacking if connected to systems, and PHI leakage and unintended memorization behaviors. Guardrail behavior should also be tested—when does it refuse vs. answer with uncertainty vs. answer confidently, and is that acceptable? How effectively is uncertainty communicated to the end user?
5. Subgroup performance: Stratify results by patient factors relevant to the use case (age, language, health literacy proxies, comorbidity complexity, etc.) and clinical setting.

After deployment, HHS should use CMS payment programs to support and incentivize healthcare institutions that establish the following ways to monitor performance and safety.

Post-Deployment Monitoring:

1. Continuous monitoring with change control: Should include track versioning and model-change documentation and re-validation should be triggered when the model updates, retrieval sources change, prompts/templates change, or workflow integration changes.
2. Real-world performance surveillance: Track errors and near-misses tied to AI outputs with CAPA proceedings, establish a pathway for clinician feedback (“this is wrong/unsafe”) that feeds governance review, and ensure reporting structure encourages error-reporting databases such as FDA’s MAUDE.
3. Drift and distribution shift monitoring programs: Monitor shifts in case mix, language use, documentation templates, seasonal patterns, new guidelines/drug approvals, and subpopulation performance. Re-test on a rolling benchmark set that reflects current practice.

4. **Auditable provenance:** Capture prompt context, model/version, retrieved sources (if any), output, and downstream action (accepted/edited/ignored) in a way that supports incident review.

Ensuring that AI used in clinical care is rigorously evaluated, continuously monitored, and accountable in real-world practice protects American patients, supports American clinicians, and prevents our healthcare system from becoming dependent on opaque technologies we cannot measure, audit, or control.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

Independent testing and certification help the strongest products rise. ECRI encourages HHS to support independent programs that evaluate AI robustness, transparency, equity of performance, and monitoring readiness—giving buyers confidence and innovators a fair playing field.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

The path that best serves Americans is simple: deploy AI where it demonstrably reduces harm or burden, prove it in real-world practice, and hold vendors and deployers accountable when performance drifts so U.S. patients don't become the test bed for opaque systems that do not have the mechanism to be monitored, audited, or controlled.

AI performs best when the task is clearly defined and aligned with user needs; outcomes are measurable, and integration is thoughtfully designed. Examples of AI working well include medical imaging applications and time- critical workflows. Medical Imaging AI, the majority of the list of FDA-cleared AI/ML-enabled devices, and health systems are using these tools to support detection/triage and workflow prioritization. The strongest value case is speed and consistency for narrow tasks (e.g., flagging suspected findings for urgent review), not replacing clinician judgment. Documentation support (AI-enabled ambient scribe use cases) is another example; ambient documentation tools are showing credible benefits in [reducing documentation burden](#) and burnout and returning time to clinicians.

However, AI struggles when models are opaque, alerts overwhelm clinicians, and performance erodes over time. Examples of areas where AI in clinical use is struggling include high-stakes clinical prediction without rigorous external validation and monitoring. A widely used proprietary [sepsis prediction model](#) (implemented at many hospitals) performed substantially worse than expected in real-world evaluation, missing many true cases and generating false alerts—illustrating the risk of deploying models that look good on paper but fail under local workflows and data patterns.

Another pain point is generalizability and “silent failure modes.” Clinical AI can [fail unevenly across populations](#) when training data and evaluation do not reflect real patients and real workflows, risking worse outcomes for Americans. LLM-enabled tools in clinical settings for non-medical device

deployments are a third area, where AI can be confidently wrong, hard to audit, and it is hard to assign accountability. Even when these tools are positioned as “assistive,” they can insert subtle bias into decisions, produce plausible but incorrect content, and change behavior after updates, without the traceability needed to investigate errors. AI-enabled non-medical devices can present legal and consent risks when workflows aren’t designed around transparency and patient permission, and nothing stifles innovation and growth like litigation.

An America-first approach to clinical AI means insisting on proof, transparency, and accountability before scaling. AI should strengthen, not replace, clinical judgment, protect patients from preventable harm, and deliver measurable value to the healthcare system Americans depend on. By prioritizing tools that perform reliably in real-world care, enforcing clear expectations when they fall short, and refusing to outsource trust to opaque technologies, we can ensure innovation serves patients, clinicians, and taxpayers—rather than the other way around.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Adoption decisions should involve clinical leaders, IT leaders, safety teams, procurement, administration, and legal counsel. Clear national expectations would help these stakeholders align around value rather than novelty.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Interoperability is a strategic asset--AI cannot outperform the data it depends on. ECRI encourages continued federal leadership in data standards and exchange, enabling U.S. innovators to build tools that work anywhere in the country, and potentially beyond. The Office of the National Coordinator for Health Information Technology (ONC) [HTI-1 final rule](#) is a meaningful step toward this goal.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

From ECRI’s perspective, patients want AI that reduces wait times, improves safety, and helps clinicians focus on care, not screens. Patients worry about loss of privacy, unfair or unexplained decisions, and errors without recourse. Public trust is not a barrier to innovation; it is a prerequisite for success at scale. For many patients, the central question is not whether AI is advanced, but whether it is safe, transparent, and accountable, and whether someone remains clearly responsible when AI influences care. An effective HHS approach should therefore ensure that AI strengthens clinical relationships, protects patient autonomy and privacy, and delivers measurable benefits patients can actually see and feel, rather than introducing new risks hidden behind technical complexity.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

a. Are there published findings about the impact of adopted AI tools and their use clinical care?

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

HHS can accelerate adoption through targeted research that validates AI's real-world clinical value, including:

- **AI Software for Improving Outpatient Scheduling and Patient Chart Management**

- AI software uses nonlinear algorithms and machine learning to perform computer-based tasks. Some nonclinical AI applications in healthcare include optimizing patient scheduling and identifying patient information and data stored in medical records.
- AI applications appear to predict patient behaviors (e.g., no-show for appointments) and length of appointments and have the potential to create efficiencies in outpatient appointment scheduling. However, available studies are highly heterogeneous for the patients they are applied to, use different AI models, and lack reporting of patient outcomes, which prevents determining the risks, benefits, and safety of using AI for scheduling and other administrative functions. Most studies lacked a comparison to other methods, so it is unclear how well AI works compared with traditional approaches.
- Large, multicenter studies that compare AI with traditional methods for patient scheduling or administrative functions and that report on patient-oriented outcomes are needed.

- **AI-based Early Warning Systems for Assessing Sepsis Risk**

- Several AI prediction tools on the market are intended to identify hospitalized patients at high risk of developing sepsis.

--TREWS: Evidence from two cohort studies suggests that TREWS can reduce sepsis mortality by enabling earlier diagnosis and treatment initiation; however, the data are too limited in quantity and quality to support conclusions. Diagnostic accuracy data are at risk of bias and need independent validation, while data from two studies comparing patients with positive and negative or dismissed TREWS alerts provide only indirect evidence of TREWS's clinical utility. Additional prospective studies are needed to confirm TREWS's accuracy and to compare sepsis outcomes in patients managed with and without TREWS.

--Immunoscore: Evidence from a diagnostic cohort study shows the Sepsis Immunoscore can predict imminent sepsis onset with moderate accuracy; however, whether this may translate to time-to-treatment, morbidity, and mortality reduction is unclear because no studies report on these outcomes. The study supports only low-confidence conclusions because no independent studies are available to validate the findings. In addition to additional diagnostic performance studies, longitudinal studies that compare sepsis management with and without ImmunoScore are needed to assess its utility.

- **AI Use in Histopathology and Cytology Practice**
 - In pathology, the digitization of glass slides into high-resolution whole-slide images (WSIs) has facilitated the application of AI algorithms to histopathology and cytology workflows. Advances in computational power, data sharing, and sophisticated AI techniques such as machine learning, deep learning, computer vision, and natural language processing have transformed the field. AI aids in the analysis of large datasets, anomaly detection, tumor grading, and workflow optimization, alleviating the burden on pathologists while enhancing diagnostic accuracy and efficiency.
 - Critical healthcare challenges also underscore the need for AI adoption in pathology. Misdiagnosis remains a major issue, leading to patient harm, disabilities, and deaths. AI can address this by providing consistent, accurate, and timely diagnostic insights, reducing errors, and complementing human expertise. AI tools are also pivotal in identifying high-risk patients, improving resource allocation, and addressing the global shortage of pathologists, exacerbated by an aging workforce and declining interest in the specialty. AI-driven solutions can automate repetitive tasks, improve diagnostic accuracy, and ensure equitable access to services, addressing workforce and infrastructure challenges.
 - Evidence from studies indicates that AI-enabled diagnostic reviews of WSI demonstrate high sensitivity, specificity, and diagnostic accuracy relative to manual review across various malignancies including gastric, breast, and prostate cancer. Data to assess AI-enabled histopathologic diagnosis of other conditions is limited. Clinical utility studies show substantial agreement between AI-assisted and manual slide reviews, suggest reduced review times, and may improve detection of missed diagnoses.
- **AI-Assisted Quantitative MRI for Monitoring Multiple Sclerosis (MS) Progression**
 - Use of MRI for monitoring MS progression is limited by lengthy protocols, error-prone procedures for identifying disease markers (e.g., lesions), and the limited predictive value of existing imaging biomarkers. Several AI-based software systems on the market may address these problems and enhance diagnostic accuracy, refine treatment approaches, and improve overall patient outcomes for MS.
 - One recent systematic review (SR) examines AI modalities for MS diagnosis based on 23 case-control studies published from 2016 to 2023, one SR assesses commercially available AI quantitative volumetric reporting tools, 2 diagnostic cohort studies compare AI-assisted radiologic reports (Pixly.Neuro.MS, iQ-Solutions MS) with conventional radiologic reports, and 5 case series use 2 different AI software packages (iQ-Solutions MS, 1 study and icobrain MS, 4 studies) to monitor MS progression.
 - The case-control SR reports high sensitivity and specificity for AI programs. In the two studies comparing AI with standard radiology reports, AI improved new lesion detection; however, the AI modalities' effect on patient outcomes is unclear. The five-case series using AI to monitor disease progression reported mixed results. Many studies rely on retrospective data collection and therefore have a high risk of bias. Commercially available AI software packages have not been adequately tested in clinical situations, so additional high-quality studies are needed to compare software packages in clinical conditions.

- **AI Software for Evaluating Coronary Artery Disease (CAD)**

- There is an emerging class of AI software applications intended to facilitate CAD diagnosis and evaluation with coronary computed tomography angiography. Many use interactive 3-D rendering of the patient's coronary vasculature, which clinicians can use to assess individual lesion risk and overall disease burden.
- Clinical literature consistently highlights that AI products for CAD improve diagnostic accuracy, streamline clinical workflows, and enhance patient risk stratification compared to traditional methods. Findings include the reduction of invasive procedures, improved mortality rates, and higher prognostic value than traditional risk scores or stenosis severity alone. However, more prospective studies reporting actual outcomes are needed to validate clinical utility and optimal place in the CAD management pathway. The evidence base also lacks long-term outcomes data, real world generalizability, and cost-effectiveness studies.

AI is actively transforming healthcare, and the potential to improve clinical outcomes, reduce costs, and minimize healthcare inequities is immense. But so too is the potential for preventable harm.

ECRI urges HHS to support an accountability-first strategy for clinical AI adoption—requiring transparency, performance evidence, and ongoing monitoring. AI should enhance, not replace, clinical judgment; protect patient safety; and deliver measurable value to the healthcare system Americans depend on. Ultimately, AI is an advanced tool to assist clinicians and healthcare staff, but human decision-making remains essential.

By prioritizing tools that are tested, proven, and trusted, HHS can ensure that U.S. innovation advances patient care with integrity, equity, and reliability.

Sincerely,

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